API Reference

This is the class and function reference of scikit-learn. Please refer to the full user guide for further details, as the raw specifications of classes and functions may not be enough to give full guidelines on their use. For reference on concepts repeated across the API, see Glossary of Common Terms and API Elements.

All entries per page Search:

Object	Description
`config_context`	Context manager to temporarily change the global `sklearn`
`get_config`	Retrieve the current scikit-learn configuration. `sklearn`
`set_config`	Set global scikit-learn configuration. `sklearn`
`show_versions`	Print useful debugging information. `sklearn`
`BaseEstimator`	Base class for all estimators in scikit-learn. `sklearn.base`
`BiclusterMixin`	Mixin class for all bicluster estimators in scikit-learn `sklearn.base`
`ClassNamePrefixFeaturesOutMixin`	Mixin class for transformers that generate their owr `sklearn.base`
`ClassifierMixin`	Mixin class for all classifiers in scikit-learn. `sklearn.base`
`ClusterMixin`	Mixin class for all cluster estimators in scikit-learn. `sklearn.base`
`DensityMixin`	Mixin class for all density estimators in scikit-learn. `sklearn.base`

Object	Description
MetaEstimatorMixin	Mixin class for all meta estimators in scikit-learn. sklearn.base
OneToOneFeatureMixin	Provides get_feature_names_out for simple transfc sklearn.base
OutlierMixin	Mixin class for all outlier detection estimators in sci sklearn.base
RegressorMixin	Mixin class for all regression estimators in scikit-lea sklearn.base
TransformerMixin	Mixin class for all transformers in scikit-learn. sklearn.base
clone	Construct a new unfitted estimator with the same p sklearn.base
is_classifier	Return True if the given estimator is (probably) a cla sklearn.base
is_clusterer	Return True if the given estimator is (probably) a clu sklearn.base
is_regressor	Return True if the given estimator is (probably) a re sklearn.base
is_outlier_detector	Return True if the given estimator is (probably) an c sklearn.base
CalibratedClassifierCV	Calibrate probabilities using isotonic, sigmoid, or te sklearn.calibration
calibration_curve	Compute true and predicted probabilities for a calil sklearn.calibration
CalibrationDisplay	Calibration curve (also known as reliability diagram) sklearn.calibration
AffinityPropagation	Perform Affinity Propagation Clustering of data. sklearn.cluster

Object	Description
AgglomerativeClustering	Agglomerative Clustering. sklearn.cluster
Birch	Implements the BIRCH clustering algorithm. sklearn.cluster
BisectingKMeans	Bisecting K-Means clustering. sklearn.cluster
DBSCAN	Perform DBSCAN clustering from vector array or dis sklearn.cluster
FeatureAgglomeration	Agglomerate features. sklearn.cluster
HDBSCAN	Cluster data using hierarchical density-based cluste sklearn.cluster
KMeans	K-Means clustering. sklearn.cluster
MeanShift	Mean shift clustering using a flat kernel. sklearn.cluster
MiniBatchKMeans	Mini-Batch K-Means clustering. sklearn.cluster
OPTICS	Estimate clustering structure from vector array. sklearn.cluster
SpectralBiclustering	Spectral biclustering (Kluger, 2003) [R2af9f5762274 sklearn.cluster
SpectralClustering	Apply clustering to a projection of the normalized L sklearn.cluster
SpectralCoclustering	Spectral Co-Clustering algorithm (Dhillon, 2001) [R(sklearn.cluster
affinity_propagation	Perform Affinity Propagation Clustering of data. sklearn.cluster

Object	Description
cluster_optics_dbscan	Perform DBSCAN extraction for an arbitrary epsilon sklearn.cluster
cluster_optics_xi	Automatically extract clusters according to the Xi-st sklearn.cluster
compute_optics_graph	Compute the OPTICS reachability graph. sklearn.cluster
dbscan	Perform DBSCAN clustering from vector array or dis sklearn.cluster
estimate_bandwidth	Estimate the bandwidth to use with the mean-shift sklearn.cluster
k_means	Perform K-means clustering algorithm. sklearn.cluster
kmeans_plusplus	Init n_clusters seeds according to k-means++. sklearn.cluster
mean_shift	Perform mean shift clustering of data using a flat ke sklearn.cluster
spectral_clustering	Apply clustering to a projection of the normalized L sklearn.cluster
ward_tree	Ward clustering based on a Feature matrix. sklearn.cluster
ColumnTransformer	Applies transformers to columns of an array or pan sklearn.compose
TransformedTargetRegressor	Meta-estimator to regress on a transformed target. sklearn.compose
make_column_selector	Create a callable to select columns to be used with sklearn.compose
make_column_transformer	Construct a ColumnTransformer from the given trar sklearn.compose

Object	Description
EllipticEnvelope	An object for detecting outliers in a Gaussian distrib sklearn.covariance
EmpiricalCovariance	Maximum likelihood covariance estimator. sklearn.covariance
GraphicalLasso	Sparse inverse covariance estimation with an l1-per sklearn.covariance
GraphicalLassoCV	Sparse inverse covariance w/ cross-validated choice sklearn.covariance
LedoitWolf	LedoitWolf Estimator. sklearn.covariance
MinCovDet	Minimum Covariance Determinant (MCD): robust e: sklearn.covariance
OAS	Oracle Approximating Shrinkage Estimator. sklearn.covariance
ShrunkCovariance	Covariance estimator with shrinkage. sklearn.covariance
empirical_covariance	Compute the Maximum likelihood covariance estin sklearn.covariance
graphical_lasso	L1-penalized covariance estimator. sklearn.covariance
ledoit_wolf	Estimate the shrunk Ledoit-Wolf covariance matrix. sklearn.covariance
ledoit_wolf_shrinkage	Estimate the shrunk Ledoit-Wolf covariance matrix. sklearn.covariance
oas	Estimate covariance with the Oracle Approximating sklearn.covariance
shrunk_covariance	Calculate covariance matrices shrunk on the diagor sklearn.covariance

Object	Description
CCA	Canonical Correlation Analysis, also known as "Moc sklearn.cross_decomposition
PLSCanonical	Partial Least Squares transformer and regressor. sklearn.cross_decomposition
PLSRegression	PLS regression. sklearn.cross_decomposition
PLSSVD	Partial Least Square SVD. sklearn.cross_decomposition
clear_data_home	Delete all the content of the data home cache. sklearn.datasets
dump_svmlight_file	Dump the dataset in svmlight / libsvm file format. sklearn.datasets
fetch_20newsgroups	Load the filenames and data from the 20 newsgrou sklearn.datasets
fetch_20newsgroups_vectorized	Load and vectorize the 20 newsgroups dataset (clas sklearn.datasets
fetch_california_housing	Load the California housing dataset (regression). sklearn.datasets
fetch_covtype	Load the covertype dataset (classification). sklearn.datasets
fetch_file	Fetch a file from the web if not already present in tl sklearn.datasets
fetch_kddcup99	Load the kddcup99 dataset (classification). sklearn.datasets
fetch_lfw_pairs	Load the Labeled Faces in the Wild (LFW) pairs data sklearn.datasets
fetch_lfw_people	Load the Labeled Faces in the Wild (LFW) people da sklearn.datasets

Object	Description
fetch_olivetti_faces	Load the Olivetti faces data-set from AT&T (classific sklearn.datasets
fetch_openml	Fetch dataset from openml by name or dataset id. sklearn.datasets
fetch_rcv1	Load the RCV1 multilabel dataset (classification). sklearn.datasets
fetch_species_distributions	Loader for species distribution dataset from Phillips sklearn.datasets
get_data_home	Return the path of the scikit-learn data directory. sklearn.datasets
load_breast_cancer	Load and return the breast cancer Wisconsin datase sklearn.datasets
load_diabetes	Load and return the diabetes dataset (regression). sklearn.datasets
load_digits	Load and return the digits dataset (classification). sklearn.datasets
load_files	Load text files with categories as subfolder names. sklearn.datasets
load_iris	Load and return the iris dataset (classification). sklearn.datasets
load_linnerud	Load and return the physical exercise Linnerud data sklearn.datasets
load_sample_image	Load the numpy array of a single sample image. sklearn.datasets
load_sample_images	Load sample images for image manipulation. sklearn.datasets
load_svmlight_file	Load datasets in the svmlight / libsvm format into s sklearn.datasets

Object	Description
load_svmlight_files	Load dataset from multiple files in SVMlight format sklearn.datasets
load_wine	Load and return the wine dataset (classification). sklearn.datasets
make_biclusters	Generate a constant block diagonal structure array sklearn.datasets
make_blobs	Generate isotropic Gaussian blobs for clustering. sklearn.datasets
make_checkerboard	Generate an array with block checkerboard structur sklearn.datasets
make_circles	Make a large circle containing a smaller circle in 2d sklearn.datasets
make_classification	Generate a random n-class classification problem. sklearn.datasets
make_friedman1	Generate the "Friedman #1" regression problem. sklearn.datasets
make_friedman2	Generate the "Friedman #2" regression problem. sklearn.datasets
make_friedman3	Generate the "Friedman #3" regression problem. sklearn.datasets
make_gaussian_quantiles	Generate isotropic Gaussian and label samples by c sklearn.datasets
make_hastie_10_2	Generate data for binary classification used in Hasti sklearn.datasets
make_low_rank_matrix	Generate a mostly low rank matrix with bell-shapec sklearn.datasets
make_moons	Make two interleaving half circles. sklearn.datasets

Object	Description
make_multilabel_classification	Generate a random multilabel classification probler sklearn.datasets
make_regression	Generate a random regression problem. sklearn.datasets
make_s_curve	Generate an S curve dataset. sklearn.datasets
make_sparse_coded_signal	Generate a signal as a sparse combination of dictio sklearn.datasets
make_sparse_spd_matrix	Generate a sparse symmetric definite positive matri sklearn.datasets
make_sparse_uncorrelated	Generate a random regression problem with sparse sklearn.datasets
make_spd_matrix	Generate a random symmetric, positive-definite ma sklearn.datasets
make_swiss_roll	Generate a swiss roll dataset. sklearn.datasets
DictionaryLearning	Dictionary learning. sklearn.decomposition
FactorAnalysis	Factor Analysis (FA). sklearn.decomposition
FastICA	FastICA: a fast algorithm for Independent Compone sklearn.decomposition
IncrementalPCA	Incremental principal components analysis (IPCA). sklearn.decomposition
KernelPCA	Kernel Principal component analysis (KPCA). sklearn.decomposition
LatentDirichletAllocation	Latent Dirichlet Allocation with online variational Ba sklearn.decomposition

Object	Description
MiniBatchDictionaryLearning	Mini-batch dictionary learning. sklearn.decomposition
MiniBatchNMF	Mini-Batch Non-Negative Matrix Factorization (NM sklearn.decomposition
MiniBatchSparsePCA	Mini-batch Sparse Principal Components Analysis. sklearn.decomposition
NMF	Non-Negative Matrix Factorization (NMF). sklearn.decomposition
PCA	Principal component analysis (PCA). sklearn.decomposition
SparseCoder	Sparse coding. sklearn.decomposition
SparsePCA	Sparse Principal Components Analysis (SparsePCA). sklearn.decomposition
TruncatedSVD	Dimensionality reduction using truncated SVD (aka sklearn.decomposition
dict_learning	Solve a dictionary learning matrix factorization prob sklearn.decomposition
dict_learning_online	Solve a dictionary learning matrix factorization prob sklearn.decomposition
fastica	Perform Fast Independent Component Analysis. sklearn.decomposition
non_negative_factorization	Compute Non-negative Matrix Factorization (NMF) sklearn.decomposition
sparse_encode	Sparse coding. sklearn.decomposition
LinearDiscriminantAnalysis	Linear Discriminant Analysis. sklearn.discriminant_analysis

Object	Description
QuadraticDiscriminantAnalysis	Quadratic Discriminant Analysis. sklearn.discriminant_analysis
DummyClassifier	DummyClassifier makes predictions that ignore the sklearn.dummy
DummyRegressor	Regressor that makes predictions using simple rule sklearn.dummy
AdaBoostClassifier	An AdaBoost classifier. sklearn.ensemble
AdaBoostRegressor	An AdaBoost regressor. sklearn.ensemble
BaggingClassifier	A Bagging classifier. sklearn.ensemble
BaggingRegressor	A Bagging regressor. sklearn.ensemble
ExtraTreesClassifier	An extra-trees classifier. sklearn.ensemble
ExtraTreesRegressor	An extra-trees regressor. sklearn.ensemble
GradientBoostingClassifier	Gradient Boosting for classification. sklearn.ensemble
GradientBoostingRegressor	Gradient Boosting for regression. sklearn.ensemble
HistGradientBoostingClassifier	Histogram-based Gradient Boosting Classification T sklearn.ensemble
HistGradientBoostingRegressor	Histogram-based Gradient Boosting Regression Tre sklearn.ensemble
IsolationForest	Isolation Forest Algorithm. sklearn.ensemble

Object	Description
RandomForestClassifier	A random forest classifier. sklearn.ensemble
RandomForestRegressor	A random forest regressor. sklearn.ensemble
RandomTreesEmbedding	An ensemble of totally random trees. sklearn.ensemble
StackingClassifier	Stack of estimators with a final classifier. sklearn.ensemble
StackingRegressor	Stack of estimators with a final regressor. sklearn.ensemble
VotingClassifier	Soft Voting/Majority Rule classifier for unfitted estir sklearn.ensemble
VotingRegressor	Prediction voting regressor for unfitted estimators. sklearn.ensemble
ConvergenceWarning	Custom warning to capture convergence problems sklearn.exceptions
DataConversionWarning	Warning used to notify implicit data conversions ha sklearn.exceptions
DataDimensionalityWarning	Custom warning to notify potential issues with data sklearn.exceptions
EfficiencyWarning	Warning used to notify the user of inefficient comp sklearn.exceptions
FitFailedWarning	Warning class used if there is an error while fitting t sklearn.exceptions
InconsistentVersionWarning	Warning raised when an estimator is unpickled witł sklearn.exceptions
NotFittedError	Exception class to raise if estimator is used before f sklearn.exceptions

Object	Description
UndefinedMetricWarning	Warning used when the metric is invalid sklearn.exceptions
EstimatorCheckFailedWarning	Warning raised when an estimator check from the c sklearn.exceptions
enable_halving_search_cv	Enables Successive Halving search-estimators sklearn.experimental
enable_iterative_imputer	Enables IterativeImputer sklearn.experimental
DictVectorizer	Transforms lists of feature-value mappings to vecto sklearn.feature_extraction
FeatureHasher	Implements feature hashing, aka the hashing trick. sklearn.feature_extraction
PatchExtractor	Extracts patches from a collection of images. sklearn.feature_extraction.image
extract_patches_2d	Reshape a 2D image into a collection of patches. sklearn.feature_extraction.image
grid_to_graph	Graph of the pixel-to-pixel connections. sklearn.feature_extraction.image
img_to_graph	Graph of the pixel-to-pixel gradient connections. sklearn.feature_extraction.image
reconstruct_from_patches_2d	Reconstruct the image from all of its patches. sklearn.feature_extraction.image
CountVectorizer	Convert a collection of text documents to a matrix o sklearn.feature_extraction.text
HashingVectorizer	Convert a collection of text documents to a matrix o sklearn.feature_extraction.text
TfidfTransformer	Transform a count matrix to a normalized tf or tf-id sklearn.feature_extraction.text

Object	Description
TfidfVectorizer	Convert a collection of raw documents to a matrix c sklearn.feature_extraction.text
GenericUnivariateSelect	Univariate feature selector with configurable strateg sklearn.feature_selection
RFE	Feature ranking with recursive feature elimination. sklearn.feature_selection
RFECV	Recursive feature elimination with cross-validation t sklearn.feature_selection
SelectFdr	Filter: Select the p-values for an estimated false disc sklearn.feature_selection
SelectFpr	Filter: Select the pvalues below alpha based on a FP sklearn.feature_selection
SelectFromModel	Meta-transformer for selecting features based on ir sklearn.feature_selection
SelectFwe	Filter: Select the p-values corresponding to Family-v sklearn.feature_selection
SelectKBest	Select features according to the k highest scores. sklearn.feature_selection
SelectPercentile	Select features according to a percentile of the high sklearn.feature_selection
SelectorMixin	Transformer mixin that performs feature selection g sklearn.feature_selection
SequentialFeatureSelector	Transformer that performs Sequential Feature Selec sklearn.feature_selection
VarianceThreshold	Feature selector that removes all low-variance featu sklearn.feature_selection
chi2	Compute chi-squared stats between each non-nega sklearn.feature_selection

Object	Description
f_classif	Compute the ANOVA F-value for the provided sam sklearn.feature_selection
f_regression	Univariate linear regression tests returning F-statist sklearn.feature_selection
mutual_info_classif	Estimate mutual information for a discrete target va sklearn.feature_selection
mutual_info_regression	Estimate mutual information for a continuous targe sklearn.feature_selection
r_regression	Compute Pearson's r for each features and the targ sklearn.feature_selection
FrozenEstimator	Estimator that wraps a fitted estimator to prevent r sklearn.frozen
GaussianProcessClassifier	Gaussian process classification (GPC) based on Lapl sklearn.gaussian_process
GaussianProcessRegressor	Gaussian process regression (GPR). sklearn.gaussian_process
CompoundKernel	Kernel which is composed of a set of other kernels. sklearn.gaussian_process.kernels
ConstantKernel	Constant kernel. sklearn.gaussian_process.kernels
DotProduct	Dot-Product kernel. sklearn.gaussian_process.kernels
ExpSineSquared	Exp-Sine-Squared kernel (aka periodic kernel). sklearn.gaussian_process.kernels
Exponentiation	The Exponentiation kernel takes one base kernel ar sklearn.gaussian_process.kernels
Hyperparameter	A kernel hyperparameter's specification in form of a sklearn.gaussian_process.kernels

Object	Description
Kernel	Base class for all kernels. sklearn.gaussian_process.kernels
Matern	Matern kernel. sklearn.gaussian_process.kernels
PairwiseKernel	Wrapper for kernels in sklearn.metrics.pairwise. sklearn.gaussian_process.kernels
Product	The `Product` kernel takes two kernels k_1 and k_2 sklearn.gaussian_process.kernels
RBF	Radial basis function kernel (aka squared-exponent sklearn.gaussian_process.kernels
RationalQuadratic	Rational Quadratic kernel. sklearn.gaussian_process.kernels
Sum	The `Sum` kernel takes two kernels k_1 and k_2 sklearn.gaussian_process.kernels
WhiteKernel	White kernel. sklearn.gaussian_process.kernels
IterativeImputer	Multivariate imputer that estimates each feature fro sklearn.impute
KNNImputer	Imputation for completing missing values using k-N sklearn.impute
MissingIndicator	Binary indicators for missing values. sklearn.impute
SimpleImputer	Univariate imputer for completing missing values w sklearn.impute
partial_dependence	Partial dependence of `features` . sklearn.inspection
permutation_importance	Permutation importance for feature evaluation [Rd9 sklearn.inspection

Object	Description
DecisionBoundaryDisplay	Decisions boundary visualization. sklearn.inspection
PartialDependenceDisplay	Partial Dependence Plot (PDP) and Individual Cond sklearn.inspection
IsotonicRegression	Isotonic regression model. sklearn.isotonic
check_increasing	Determine whether y is monotonically correlated w sklearn.isotonic
isotonic_regression	Solve the isotonic regression model. sklearn.isotonic
AdditiveChi2Sampler	Approximate feature map for additive chi2 kernel. sklearn.kernel_approximation
Nystroem	Approximate a kernel map using a subset of the tra sklearn.kernel_approximation
PolynomialCountSketch	Polynomial kernel approximation via Tensor Sketch. sklearn.kernel_approximation
RBFSampler	Approximate a RBF kernel feature map using rando sklearn.kernel_approximation
SkewedChi2Sampler	Approximate feature map for "skewed chi-squared" sklearn.kernel_approximation
KernelRidge	Kernel ridge regression. sklearn.kernel_ridge
LogisticRegression	Logistic Regression (aka logit, MaxEnt) classifier. sklearn.linear_model
LogisticRegressionCV	Logistic Regression CV (aka logit, MaxEnt) classifier. sklearn.linear_model
PassiveAggressiveClassifier	Passive Aggressive Classifier. sklearn.linear_model

Object	Description
Perceptron	Linear perceptron classifier. sklearn.linear_model
RidgeClassifier	Classifier using Ridge regression. sklearn.linear_model
RidgeClassifierCV	Ridge classifier with built-in cross-validation. sklearn.linear_model
SGDClassifier	Linear classifiers (SVM, logistic regression, etc.) with sklearn.linear_model
SGDOneClassSVM	Solves linear One-Class SVM using Stochastic Gradi sklearn.linear_model
LinearRegression	Ordinary least squares Linear Regression. sklearn.linear_model
Ridge	Linear least squares with l2 regularization. sklearn.linear_model
RidgeCV	Ridge regression with built-in cross-validation. sklearn.linear_model
SGDRegressor	Linear model fitted by minimizing a regularized em sklearn.linear_model
ElasticNet	Linear regression with combined L1 and L2 priors a sklearn.linear_model
ElasticNetCV	Elastic Net model with iterative fitting along a regu sklearn.linear_model
Lars	Least Angle Regression model a.k.a. LAR. sklearn.linear_model
LarsCV	Cross-validated Least Angle Regression model. sklearn.linear_model
Lasso	Linear Model trained with L1 prior as regularizer (ak sklearn.linear_model

Object	Description
LassoCV	Lasso linear model with iterative fitting along a regu sklearn.linear_model
LassoLars	Lasso model fit with Least Angle Regression a.k.a. L sklearn.linear_model
LassoLarsCV	Cross-validated Lasso, using the LARS algorithm. sklearn.linear_model
LassoLarsIC	Lasso model fit with Lars using BIC or AIC for mode sklearn.linear_model
OrthogonalMatchingPursuit	Orthogonal Matching Pursuit model (OMP). sklearn.linear_model
OrthogonalMatchingPursuitCV	Cross-validated Orthogonal Matching Pursuit mode sklearn.linear_model
ARDRegression	Bayesian ARD regression. sklearn.linear_model
BayesianRidge	Bayesian ridge regression. sklearn.linear_model
MultiTaskElasticNet	Multi-task ElasticNet model trained with L1/L2 mixe sklearn.linear_model
MultiTaskElasticNetCV	Multi-task L1/L2 ElasticNet with built-in cross-valida sklearn.linear_model
MultiTaskLasso	Multi-task Lasso model trained with L1/L2 mixed-n sklearn.linear_model
MultiTaskLassoCV	Multi-task Lasso model trained with L1/L2 mixed-n sklearn.linear_model
HuberRegressor	L2-regularized linear regression model that is robus sklearn.linear_model
QuantileRegressor	Linear regression model that predicts conditional q sklearn.linear_model

Object	Description
RANSACRegressor	RANSAC (RANdom SAmple Consensus) algorithm. sklearn.linear_model
TheilSenRegressor	Theil-Sen Estimator: robust multivariate regression sklearn.linear_model
GammaRegressor	Generalized Linear Model with a Gamma distributic sklearn.linear_model
PoissonRegressor	Generalized Linear Model with a Poisson distributio sklearn.linear_model
TweedieRegressor	Generalized Linear Model with a Tweedie distributic sklearn.linear_model
PassiveAggressiveRegressor	Passive Aggressive Regressor. sklearn.linear_model
enet_path	Compute elastic net path with coordinate descent. sklearn.linear_model
lars_path	Compute Least Angle Regression or Lasso path usir sklearn.linear_model
lars_path_gram	The lars_path in the sufficient stats mode. sklearn.linear_model
lasso_path	Compute Lasso path with coordinate descent. sklearn.linear_model
orthogonal_mp	Orthogonal Matching Pursuit (OMP). sklearn.linear_model
orthogonal_mp_gram	Gram Orthogonal Matching Pursuit (OMP). sklearn.linear_model
ridge_regression	Solve the ridge equation by the method of normal sklearn.linear_model
ClassicalMDS	Classical multidimensional scaling (MDS). sklearn.manifold

Object	Description
Isomap	Isomap Embedding. sklearn.manifold
LocallyLinearEmbedding	Locally Linear Embedding. sklearn.manifold
MDS	Multidimensional scaling. sklearn.manifold
SpectralEmbedding	Spectral embedding for non-linear dimensionality r sklearn.manifold
TSNE	T-distributed Stochastic Neighbor Embedding. sklearn.manifold
locally_linear_embedding	Perform a Locally Linear Embedding analysis on the sklearn.manifold
smacof	Compute multidimensional scaling using the SMAC sklearn.manifold
spectral_embedding	Project the sample on the first eigenvectors of the sklearn.manifold
trustworthiness	Indicate to what extent the local structure is retaine sklearn.manifold
check_scoring	Determine scorer from user options. sklearn.metrics
get_scorer	Get a scorer from string. sklearn.metrics
get_scorer_names	Get the names of all available scorers. sklearn.metrics
make_scorer	Make a scorer from a performance metric or loss fu sklearn.metrics
accuracy_score	Accuracy classification score. sklearn.metrics

Object	Description
auc	Compute Area Under the Curve (AUC) using the tra sklearn.metrics
average_precision_score	Compute average precision (AP) from prediction sc sklearn.metrics
balanced_accuracy_score	Compute the balanced accuracy. sklearn.metrics
brier_score_loss	Compute the Brier score loss. sklearn.metrics
class_likelihood_ratios	Compute binary classification positive and negative sklearn.metrics
classification_report	Build a text report showing the main classification r sklearn.metrics
cohen_kappa_score	Compute Cohen's kappa: a statistic that measures i sklearn.metrics
confusion_matrix	Compute confusion matrix to evaluate the accuracy sklearn.metrics
confusion_matrix_at_thresholds	Calculate binary confusion matrix terms per classifi sklearn.metrics
d2_brier_score	D^2 score function, fraction of Brier score explained. sklearn.metrics
d2_log_loss_score	D^2 score function, fraction of log loss explained. sklearn.metrics
dcg_score	Compute Discounted Cumulative Gain. sklearn.metrics
det_curve	Compute Detection Error Tradeoff (DET) for differer sklearn.metrics
f1_score	Compute the F1 score, also known as balanced F-sc sklearn.metrics

Object	Description
fbeta_score	Compute the F-beta score. sklearn.metrics
hamming_loss	Compute the average Hamming loss. sklearn.metrics
hinge_loss	Average hinge loss (non-regularized). sklearn.metrics
jaccard_score	Jaccard similarity coefficient score. sklearn.metrics
log_loss	Log loss, aka logistic loss or cross-entropy loss. sklearn.metrics
matthews_corrcoef	Compute the Matthews correlation coefficient (MC sklearn.metrics
multilabel_confusion_matrix	Compute a confusion matrix for each class or samp sklearn.metrics
ndcg_score	Compute Normalized Discounted Cumulative Gain. sklearn.metrics
precision_recall_curve	Compute precision-recall pairs for different probab sklearn.metrics
precision_recall_fscore_support	Compute precision, recall, F-measure and support f sklearn.metrics
precision_score	Compute the precision. sklearn.metrics
recall_score	Compute the recall. sklearn.metrics
roc_auc_score	Compute Area Under the Receiver Operating Chara from prediction scores. sklearn.metrics

Object	Description
roc_curve	Compute Receiver operating characteristic (ROC). sklearn.metrics
top_k_accuracy_score	Top-k Accuracy classification score. sklearn.metrics
zero_one_loss	Zero-one classification loss. sklearn.metrics
d2_absolute_error_score	D^2 regression score function, fraction of absolute e sklearn.metrics
d2_pinball_score	D^2 regression score function, fraction of pinball los sklearn.metrics
d2_tweedie_score	D^2 regression score function, fraction of Tweedie d sklearn.metrics
explained_variance_score	Explained variance regression score function. sklearn.metrics
max_error	The max_error metric calculates the maximum resid sklearn.metrics
mean_absolute_error	Mean absolute error regression loss. sklearn.metrics
mean_absolute_percentage_error	Mean absolute percentage error (MAPE) regression sklearn.metrics
mean_gamma_deviance	Mean Gamma deviance regression loss. sklearn.metrics
mean_pinball_loss	Pinball loss for quantile regression. sklearn.metrics
mean_poisson_deviance	Mean Poisson deviance regression loss. sklearn.metrics
mean_squared_error	Mean squared error regression loss. sklearn.metrics

Object	Description
mean_squared_log_error	Mean squared logarithmic error regression loss. sklearn.metrics
mean_tweedie_deviance	Mean Tweedie deviance regression loss. sklearn.metrics
median_absolute_error	Median absolute error regression loss. sklearn.metrics
r2_score	R^2 (coefficient of determination) regression score f sklearn.metrics
root_mean_squared_error	Root mean squared error regression loss. sklearn.metrics
root_mean_squared_log_error	Root mean squared logarithmic error regression los sklearn.metrics
coverage_error	Coverage error measure. sklearn.metrics
label_ranking_average_precision_score	Compute ranking-based average precision. sklearn.metrics
label_ranking_loss	Compute Ranking loss measure. sklearn.metrics
adjusted_mutual_info_score	Adjusted Mutual Information between two clusterir sklearn.metrics
adjusted_rand_score	Rand index adjusted for chance. sklearn.metrics
calinski_harabasz_score	Compute the Calinski and Harabasz score. sklearn.metrics
contingency_matrix	Build a contingency matrix describing the relationsl sklearn.metrics.cluster
pair_confusion_matrix	Pair confusion matrix arising from two clusterings. sklearn.metrics.cluster

Object	Description
completeness_score	Compute completeness metric of a cluster labeling sklearn.metrics
davies_bouldin_score	Compute the Davies-Bouldin score. sklearn.metrics
fowlkes_mallows_score	Measure the similarity of two clusterings of a set of sklearn.metrics
homogeneity_completeness_v_measure	Compute the homogeneity and completeness and sklearn.metrics
homogeneity_score	Homogeneity metric of a cluster labeling given a gr sklearn.metrics
mutual_info_score	Mutual Information between two clusterings. sklearn.metrics
normalized_mutual_info_score	Normalized Mutual Information between two cluste sklearn.metrics
rand_score	Rand index. sklearn.metrics
silhouette_samples	Compute the Silhouette Coefficient for each sample sklearn.metrics
silhouette_score	Compute the mean Silhouette Coefficient of all sar sklearn.metrics
v_measure_score	V-measure cluster labeling given a ground truth. sklearn.metrics
consensus_score	The similarity of two sets of biclusters. sklearn.metrics
DistanceMetric	Uniform interface for fast distance metric functions sklearn.metrics
additive_chi2_kernel	Compute the additive chi-squared kernel between sklearn.metrics.pairwise

Object	Description
chi2_kernel	Compute the exponential chi-squared kernel betwe sklearn.metrics.pairwise
cosine_distances	Compute cosine distance between samples in X anc sklearn.metrics.pairwise
cosine_similarity	Compute cosine similarity between samples in X an sklearn.metrics.pairwise
distance_metrics	Valid metrics for pairwise_distances. sklearn.metrics.pairwise
euclidean_distances	Compute the distance matrix between each pair frc sklearn.metrics.pairwise
haversine_distances	Compute the Haversine distance between samples sklearn.metrics.pairwise
kernel_metrics	Valid metrics for pairwise_kernels. sklearn.metrics.pairwise
laplacian_kernel	Compute the laplacian kernel between X and Y. sklearn.metrics.pairwise
linear_kernel	Compute the linear kernel between X and Y. sklearn.metrics.pairwise
manhattan_distances	Compute the L1 distances between the vectors in X sklearn.metrics.pairwise
nan_euclidean_distances	Calculate the euclidean distances in the presence o sklearn.metrics.pairwise
paired_cosine_distances	Compute the paired cosine distances between X an sklearn.metrics.pairwise
paired_distances	Compute the paired distances between X and Y. sklearn.metrics.pairwise
paired_euclidean_distances	Compute the paired euclidean distances between X sklearn.metrics.pairwise

Object	Description
paired_manhattan_distances	Compute the paired L1 distances between X and Y. sklearn.metrics.pairwise
pairwise_kernels	Compute the kernel between arrays X and optional sklearn.metrics.pairwise
polynomial_kernel	Compute the polynomial kernel between X and Y. sklearn.metrics.pairwise
rbf_kernel	Compute the rbf (gaussian) kernel between X and Y sklearn.metrics.pairwise
sigmoid_kernel	Compute the sigmoid kernel between X and Y. sklearn.metrics.pairwise
pairwise_distances	Compute the distance matrix from a feature array X sklearn.metrics
pairwise_distances_argmin	Compute minimum distances between one point ar sklearn.metrics
pairwise_distances_argmin_min	Compute minimum distances between one point ar sklearn.metrics
pairwise_distances_chunked	Generate a distance matrix chunk by chunk with op sklearn.metrics
ConfusionMatrixDisplay	Confusion Matrix visualization. sklearn.metrics
DetCurveDisplay	Detection Error Tradeoff (DET) curve visualization. sklearn.metrics
PrecisionRecallDisplay	Precision Recall visualization. sklearn.metrics
PredictionErrorDisplay	Visualization of the prediction error of a regression sklearn.metrics
RocCurveDisplay	ROC Curve visualization. sklearn.metrics

Object	Description
BayesianGaussianMixture	Variational Bayesian estimation of a Gaussian mixtu sklearn.mixture
GaussianMixture	Gaussian Mixture. sklearn.mixture
GroupKFold	K-fold iterator variant with non-overlapping groups sklearn.model_selection
GroupShuffleSplit	Shuffle-Group(s)-Out cross-validation iterator. sklearn.model_selection
KFold	K-Fold cross-validator. sklearn.model_selection
LeaveOneGroupOut	Leave One Group Out cross-validator. sklearn.model_selection
LeaveOneOut	Leave-One-Out cross-validator. sklearn.model_selection
LeavePGroupsOut	Leave P Group(s) Out cross-validator. sklearn.model_selection
LeavePOut	Leave-P-Out cross-validator. sklearn.model_selection
PredefinedSplit	Predefined split cross-validator. sklearn.model_selection
RepeatedKFold	Repeated K-Fold cross validator. sklearn.model_selection
RepeatedStratifiedKFold	Repeated class-wise stratified K-Fold cross validato sklearn.model_selection
ShuffleSplit	Random permutation cross-validator. sklearn.model_selection
StratifiedGroupKFold	Class-wise stratified K-Fold iterator variant with nor sklearn.model_selection

Object	Description
StratifiedKFold	Class-wise stratified K-Fold cross-validator. sklearn.model_selection
StratifiedShuffleSplit	Class-wise stratified ShuffleSplit cross-validator. sklearn.model_selection
TimeSeriesSplit	Time Series cross-validator. sklearn.model_selection
check_cv	Input checker utility for building a cross-validator. sklearn.model_selection
train_test_split	Split arrays or matrices into random train and test s sklearn.model_selection
GridSearchCV	Exhaustive search over specified parameter values f sklearn.model_selection
HalvingGridSearchCV	Search over specified parameter values with succes sklearn.model_selection
HalvingRandomSearchCV	Randomized search on hyper parameters. sklearn.model_selection
ParameterGrid	Grid of parameters with a discrete number of value sklearn.model_selection
ParameterSampler	Generator on parameters sampled from given distri sklearn.model_selection
RandomizedSearchCV	Randomized search on hyper parameters. sklearn.model_selection
FixedThresholdClassifier	Binary classifier that manually sets the decision thre sklearn.model_selection
TunedThresholdClassifierCV	Classifier that post-tunes the decision threshold usi sklearn.model_selection
cross_val_predict	Generate cross-validated estimates for each input c sklearn.model_selection

Object	Description
cross_val_score	Evaluate a score by cross-validation. sklearn.model_selection
cross_validate	Evaluate metric(s) by cross-validation and also reco sklearn.model_selection
learning_curve	Learning curve. sklearn.model_selection
permutation_test_score	Evaluate the significance of a cross-validated score sklearn.model_selection
validation_curve	Validation curve. sklearn.model_selection
LearningCurveDisplay	Learning Curve visualization. sklearn.model_selection
ValidationCurveDisplay	Validation Curve visualization. sklearn.model_selection
OneVsOneClassifier	One-vs-one multiclass strategy. sklearn.multiclass
OneVsRestClassifier	One-vs-the-rest (OvR) multiclass strategy. sklearn.multiclass
OutputCodeClassifier	(Error-Correcting) Output-Code multiclass strategy. sklearn.multiclass
ClassifierChain	A multi-label model that arranges binary classifiers sklearn.multioutput
MultiOutputClassifier	Multi target classification. sklearn.multioutput
MultiOutputRegressor	Multi target regression. sklearn.multioutput
RegressorChain	A multi-label model that arranges regressions into sklearn.multioutput

Object	Description
BernoulliNB	Naive Bayes classifier for multivariate Bernoulli moc sklearn.naive_bayes
CategoricalNB	Naive Bayes classifier for categorical features. sklearn.naive_bayes
ComplementNB	The Complement Naive Bayes classifier described ir sklearn.naive_bayes
GaussianNB	Gaussian Naive Bayes (GaussianNB). sklearn.naive_bayes
MultinomialNB	Naive Bayes classifier for multinomial models. sklearn.naive_bayes
BallTree	BallTree for fast generalized N-point problems sklearn.neighbors
KDTree	KDTree for fast generalized N-point problems sklearn.neighbors
KNeighborsClassifier	Classifier implementing the k-nearest neighbors vo sklearn.neighbors
KNeighborsRegressor	Regression based on k-nearest neighbors. sklearn.neighbors
KNeighborsTransformer	Transform X into a (weighted) graph of k nearest ne sklearn.neighbors
KernelDensity	Kernel Density Estimation. sklearn.neighbors
LocalOutlierFactor	Unsupervised Outlier Detection using the Local Out sklearn.neighbors
NearestCentroid	Nearest centroid classifier. sklearn.neighbors
NearestNeighbors	Unsupervised learner for implementing neighbor se sklearn.neighbors

Object	Description
NeighborhoodComponentsAnalysis	Neighborhood Components Analysis. sklearn.neighbors
RadiusNeighborsClassifier	Classifier implementing a vote among neighbors wi sklearn.neighbors
RadiusNeighborsRegressor	Regression based on neighbors within a fixed radiu sklearn.neighbors
RadiusNeighborsTransformer	Transform X into a (weighted) graph of neighbors n sklearn.neighbors
kneighbors_graph	Compute the (weighted) graph of k-Neighbors for sklearn.neighbors
radius_neighbors_graph	Compute the (weighted) graph of Neighbors for pc sklearn.neighbors
sort_graph_by_row_values	Sort a sparse graph such that each row is stored wi sklearn.neighbors
BernoulliRBM	Bernoulli Restricted Boltzmann Machine (RBM). sklearn.neural_network
MLPClassifier	Multi-layer Perceptron classifier. sklearn.neural_network
MLPRegressor	Multi-layer Perceptron regressor. sklearn.neural_network
FeatureUnion	Concatenates results of multiple transformer object sklearn.pipeline
Pipeline	A sequence of data transformers with an optional fi sklearn.pipeline
make_pipeline	Construct a Pipeline from the given estimators. sklearn.pipeline
make_union	Construct a FeatureUnion from the given transforn sklearn.pipeline

Object	Description
Binarizer	Binarize data (set feature values to 0 or 1) accordin sklearn.preprocessing
FunctionTransformer	Constructs a transformer from an arbitrary callable. sklearn.preprocessing
KBinsDiscretizer	Bin continuous data into intervals. sklearn.preprocessing
KernelCenterer	Center an arbitrary kernel matrix K. sklearn.preprocessing
LabelBinarizer	Binarize labels in a one-vs-all fashion. sklearn.preprocessing
LabelEncoder	Encode target labels with value between 0 and n_cl sklearn.preprocessing
MaxAbsScaler	Scale each feature by its maximum absolute value. sklearn.preprocessing
MinMaxScaler	Transform features by scaling each feature to a give sklearn.preprocessing
MultiLabelBinarizer	Transform between iterable of iterables and a multi sklearn.preprocessing
Normalizer	Normalize samples individually to unit norm. sklearn.preprocessing
OneHotEncoder	Encode categorical features as a one-hot numeric a sklearn.preprocessing
OrdinalEncoder	Encode categorical features as an integer array. sklearn.preprocessing
PolynomialFeatures	Generate polynomial and interaction features. sklearn.preprocessing
PowerTransformer	Apply a power transform featurewise to make data sklearn.preprocessing

Object	Description
QuantileTransformer	Transform features using quantiles information. sklearn.preprocessing
RobustScaler	Scale features using statistics that are robust to out sklearn.preprocessing
SplineTransformer	Generate univariate B-spline bases for features. sklearn.preprocessing
StandardScaler	Standardize features by removing the mean and sc sklearn.preprocessing
TargetEncoder	Target Encoder for regression and classification targ sklearn.preprocessing
add_dummy_feature	Augment dataset with an additional dummy feature sklearn.preprocessing
binarize	Boolean thresholding of array-like or scipy.sparse m sklearn.preprocessing
label_binarize	Binarize labels in a one-vs-all fashion. sklearn.preprocessing
maxabs_scale	Scale each feature to the [-1, 1] range without brea sklearn.preprocessing
minmax_scale	Transform features by scaling each feature to a give sklearn.preprocessing
normalize	Scale input vectors individually to unit norm (vecto sklearn.preprocessing
power_transform	Parametric, monotonic transformation to make dat sklearn.preprocessing
quantile_transform	Transform features using quantiles information. sklearn.preprocessing
robust_scale	Standardize a dataset along any axis. sklearn.preprocessing

Object	Description
scale	Standardize a dataset along any axis. sklearn.preprocessing
GaussianRandomProjection	Reduce dimensionality through Gaussian random p sklearn.random_projection
SparseRandomProjection	Reduce dimensionality through sparse random proj sklearn.random_projection
johnson_lindenstrauss_min_dim	Find a 'safe' number of components to randomly pı sklearn.random_projection
LabelPropagation	Label Propagation classifier. sklearn.semi_supervised
LabelSpreading	LabelSpreading model for semi-supervised learning sklearn.semi_supervised
SelfTrainingClassifier	Self-training classifier. sklearn.semi_supervised
LinearSVC	Linear Support Vector Classification. sklearn.svm
LinearSVR	Linear Support Vector Regression. sklearn.svm
NuSVC	Nu-Support Vector Classification. sklearn.svm
NuSVR	Nu Support Vector Regression. sklearn.svm
OneClassSVM	Unsupervised Outlier Detection. sklearn.svm
SVC	C-Support Vector Classification. sklearn.svm
SVR	Epsilon-Support Vector Regression. sklearn.svm

Object	Description
l1_min_c	Return the lowest bound for C. sklearn.svm
DecisionTreeClassifier	A decision tree classifier. sklearn.tree
DecisionTreeRegressor	A decision tree regressor. sklearn.tree
ExtraTreeClassifier	An extremely randomized tree classifier. sklearn.tree
ExtraTreeRegressor	An extremely randomized tree regressor. sklearn.tree
export_graphviz	Export a decision tree in DOT format. sklearn.tree
export_text	Build a text report showing the rules of a decision t sklearn.tree
plot_tree	Plot a decision tree. sklearn.tree
Bunch	Container object exposing keys as attributes. sklearn.utils
_safe_indexing	Return rows, items or columns of X using indices. sklearn.utils
as_float_array	Convert an array-like to an array of floats. sklearn.utils
assert_all_finite	Throw a ValueError if X contains NaN or infinity. sklearn.utils
deprecated	Decorator to mark a function or class as deprecatec sklearn.utils
estimator_html_repr	Build a HTML representation of an estimator. sklearn.utils

Object	Description
gen_batches	Generator to create slices containing batch_size e sklearn.utils
gen_even_slices	Generator to create n_packs evenly spaced slices g sklearn.utils
indexable	Make arrays indexable for cross-validation. sklearn.utils
murmurhash3_32	Compute the 32bit murmurhash3 of key at seed. sklearn.utils
resample	Resample arrays or sparse matrices in a consistent w sklearn.utils
safe_mask	Return a mask which is safe to use on X. sklearn.utils
safe_sqr	Element wise squaring of array-likes and sparse ma sklearn.utils
shuffle	Shuffle arrays or sparse matrices in a consistent way sklearn.utils
Tags	Tags for the estimator. sklearn.utils
InputTags	Tags for the input data. sklearn.utils
TargetTags	Tags for the target data. sklearn.utils
ClassifierTags	Tags for the classifier. sklearn.utils
RegressorTags	Tags for the regressor. sklearn.utils
TransformerTags	Tags for the transformer. sklearn.utils

Object	Description
get_tags	Get estimator tags. sklearn.utils
check_X_y	Input validation for standard estimators. sklearn.utils
check_array	Input validation on an array, list, sparse matrix or si sklearn.utils
check_consistent_length	Check that all arrays have consistent first dimensior sklearn.utils
check_random_state	Turn seed into an np.random.RandomState instanc sklearn.utils
check_scalar	Validate scalar parameters type and value. sklearn.utils
check_is_fitted	Perform is_fitted validation for estimator. sklearn.utils.validation
check_memory	Check that `memory` is joblib.Memory-like. sklearn.utils.validation
check_symmetric	Make sure that array is 2D, square and symmetric. sklearn.utils.validation
column_or_1d	Ravel column or 1d numpy array, else raises an errc sklearn.utils.validation
has_fit_parameter	Check whether the estimator's fit method supports sklearn.utils.validation
validate_data	Validate input data and set or check feature names sklearn.utils.validation
available_if	An attribute that is available only if check returns a